

Determination of the Velocity of Sound in Air

Aim:- To determine the velocity of the sound in air using CRO

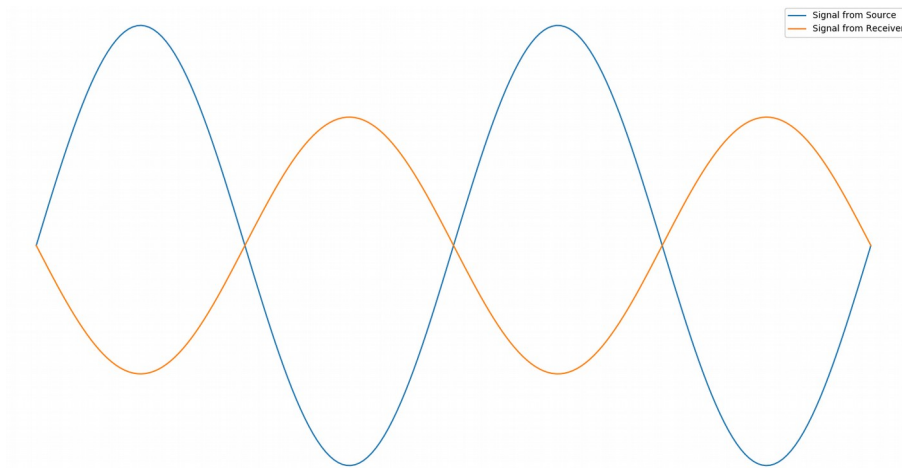
Apparatus:

- CRO
- Piezoelectric Buzzer as the source of sound
- Function Generator to produce frequencies to run the buzzer (in 2000 Hz- 6000Hz range)
- MAX4466 Electret Microphone(condenser mic) with Amplifier
- DC power supply or battery (3.3V or 5V)

MAX4466 Electret Microphone Amplifier is a dedicated amplifier chip designed specifically for microphones. It has an adjustable gain potentiometer on the back. You can turn it up to make the waveform fill the CRO screen. It is very low noise.

Principle:

When the signal from the source (Piezoelectric Buzzer) and the signal from the receiver (the condenser mic) show a phase difference of 180° , the distance between the source and receiver will be $d = \lambda/2$



$\therefore$ The wavelength of the sound waves $\lambda = 2d$. If v is the frequency of sound waves
The velocity of sound waves $V = v\lambda = 2vd$

Procedure:**Connecting the Source:**

Connect the terminal of the buzzer to the function generator and the X-Channel of the CRO . Set an audio frequency ν in the generator

Connecting the Receiver:

MAX4466 Electret Microphone Amplifier has three terminals



- VCC: Connect to 3.3V or 5V (from a DC supply or battery).
- GND: Connect to power supply ground and the CRO ground (black clip).
- OUT: Connect to the CRO probe tip (red clip).

Adjust the distance d between the source(buzzer) and the receiver (mic) to get 180° phase difference between the two signals on CRO. Note down the values of ν and d .

Repeat the experiment by varying the frequency ν and obtain the corresponding value of d

Observations:

S.No	Frequency of the Source ν (Hz)	Distance between the source and receiver d (m)	Velocity of Sound $V=2\nu d$

Mean $V = \dots\dots\dots$ m/s

Results:

The velocity of the sound in air = $\dots\dots\dots$ m/s